

MECHANICS OF PULMONARY VENTILATION

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Outlines of presentation

- ◆ **Respiratory movements**
- ◆ **Respiratory pressures**
- ◆ **Compliance of the lungs**
- ◆ **Pulmonary Function Tests**

Introduction

- ◆ Respiration is the process by which oxygen is taken in and carbon dioxide is given out
- ◆ The first breath takes place only after birth
- ◆ Permanent stoppage of respiration occurs only at death

Functional Anatomy of respiratory tract

1. Conducting Zone

✓ Nose, Pharynx, Larynx, Trachea, bronchi

2. Respiratory Zone

✓ Respiratory bronchioles, Alveolar ducts,
Alveolar Sacs , Alveoli

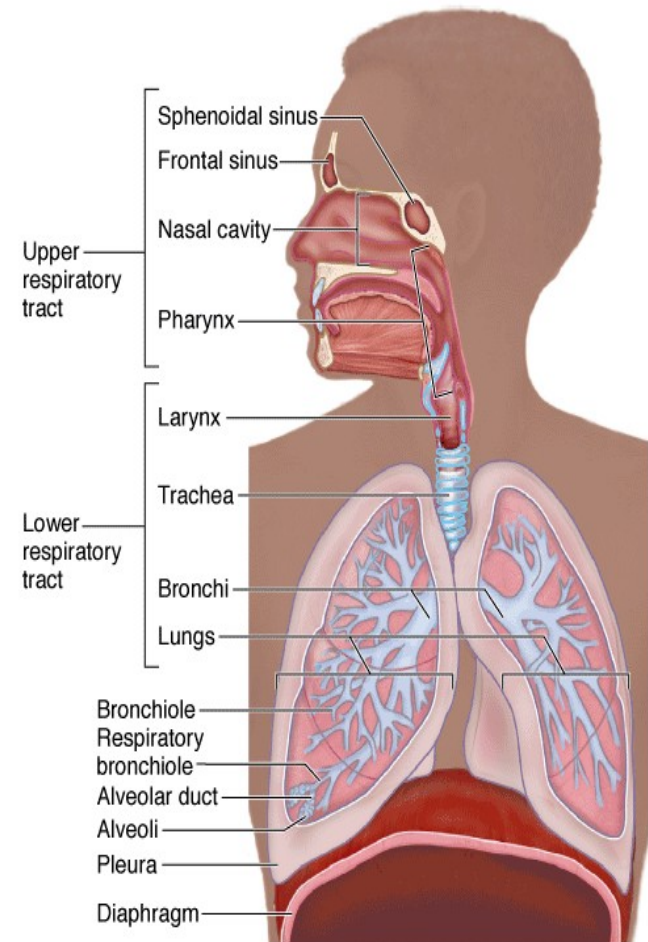
◆ Generally, respiratory tract is divided into
two parts:

i. Upper respiratory tract

✓ nose up to vocal cords; vocal cords

ii. Lower respiratory tract

✓ includes trachea, bronchi and lungs.



MECHANICS OF PULMONARY VENTILATION

1. RESPIRATORY MOVEMENTS

The lungs can be expanded and contracted in two ways:

i. by downward and upward movement of the *diaphragm*

✓ to lengthen or shorten the chest cavity

ii. by elevation and depression of the *ribs*

✓ to increase and decrease the antero-posterior diameter of the chest cavity.

Muscles of Inspiration

i. Primary inspiratory muscles

- ✓ Diaphragm

ii. Accessory inspiratory muscles

- ✓ *External intercostals*
- ✓ *Sternocleidomastoid*-lift up sternum
- ✓ *Scalene*-lift the first two ribs.
- ✓ *Anterior serrati*-lift many of the ribs
- ✓ Elevators of scapulae and pectorals

Muscles of Expiration

i. Primary expiratory muscles

- ✓ internal intercostal muscles

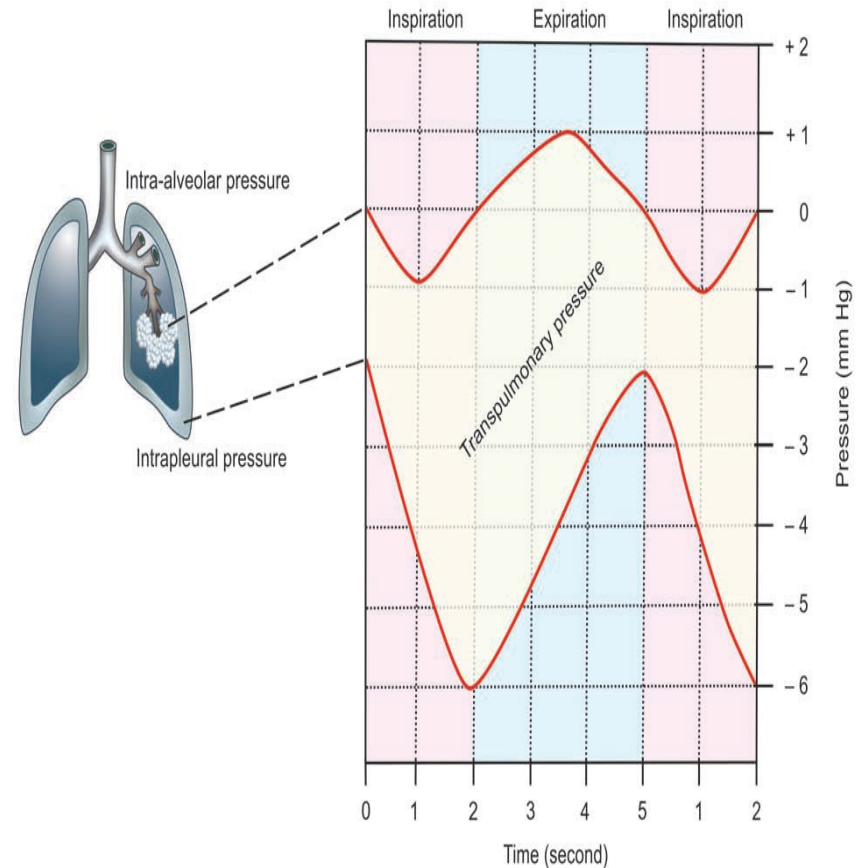
ii. Accessory expiratory muscles

- ✓ abdominal muscles

Respiratory pressures

i. Alveolar Pressure (Intra-alveolar pressure)

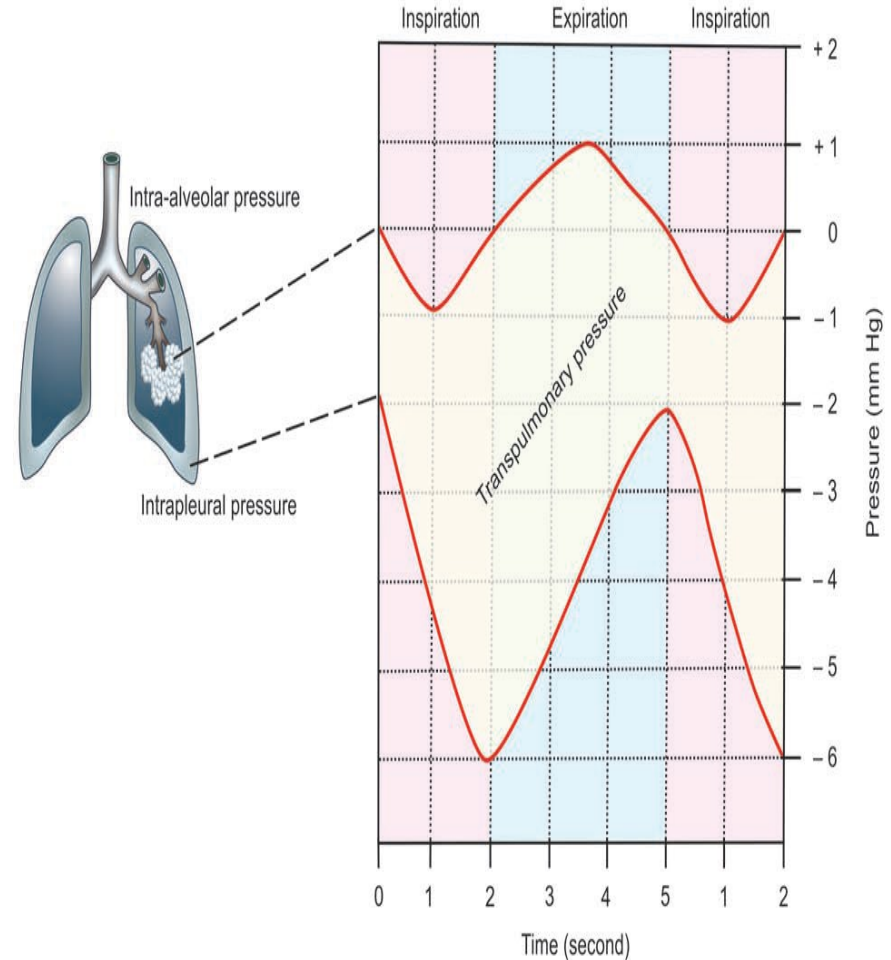
1. During normal inspiration: **-1 mm Hg**
2. During normal expiration: **+1 mm Hg**
3. At the end of inspiration and expiration: Equal to atmospheric pressure (**0 mm Hg**)
4. During forced inspiration with closed glottis (Müller maneuver): **-80 mm Hg**
5. During forced expiration with closed glottis (Valsalva maneuver): **+100 mm**



Respiratory pressures

ii. Pleural Pressure (Intra-pleural pressure)

1. At the end of normal inspiration: **−6 mm Hg**
2. At the end of normal expiration: **−2 mm Hg**
3. At the end of forced inspiration: **−30 mm Hg**
4. At the end of forced inspiration with closed glottis (Müller maneuver): **−70 mm Hg**
5. At the end of forced expiration with closed glottis (Valsalva maneuver): **+50 mm Hg**.



Q1: Why Intrapleural pressure is always −ve ?

Q2: What happens during stab wound to chest?

Respiratory pressures

- **Intra-pleural pressure functions:**
 - i. It prevents the collapsing tendency of lungs
 - ii. Because of the negative pressure in thoracic region, larger veins and vena cava are enlarged (dilated)
- ✓ Also, the negative pressure acts like suction pump and pulls the venous blood from lower part of body towards the heart against gravity. (*i.e. why Severe COPD (Pleuria problem): Ankle edema*)
- ✓ Thus, the intra-pleural pressure is responsible for venous return. So, it is called the **respiratory pump for venous return**

Q3: Why intra-pleural pressure is a.k.a intra-thoracic pressure?

Q4: How can we practically measure Intra-pleural pressure?

Respiratory pressures

iii. Trans-pulmonary Pressure

- Trans-pulmonary pressure is the pressure difference between intra-alveolar pressure and intra-pleural pressure.
- It is the measure of *elastic forces* in lungs, which is responsible for collapsing tendency of lungs.
- Always positive

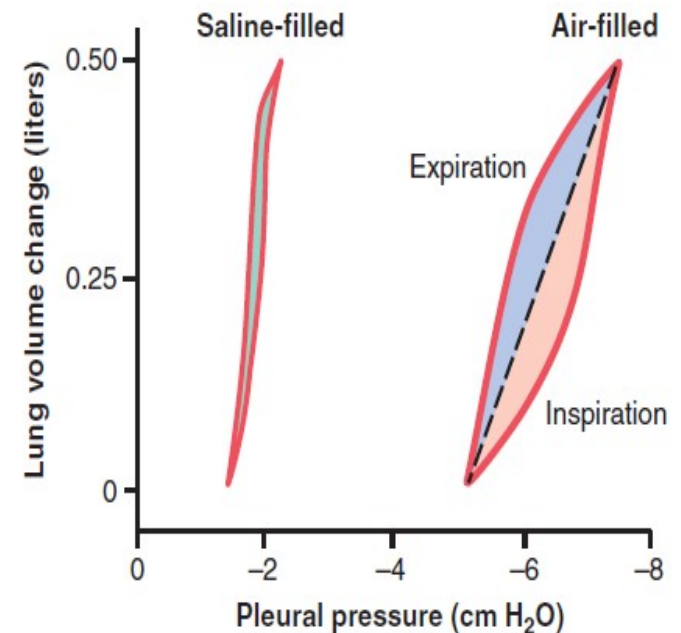
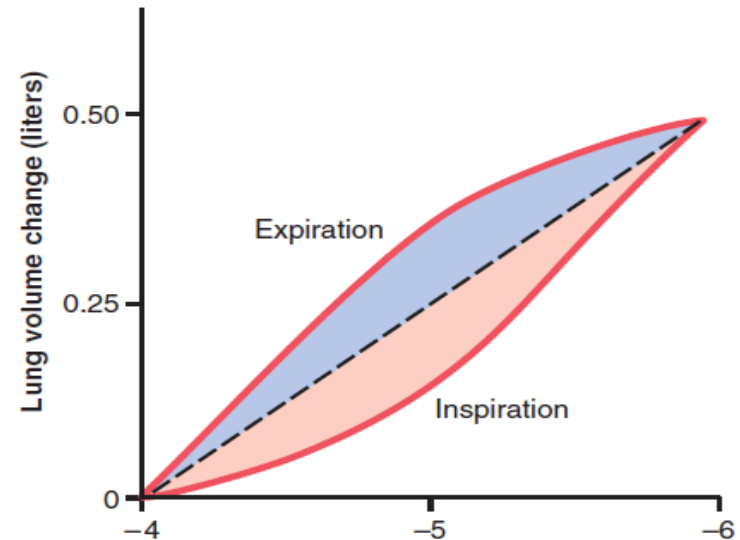
COMPLIANCE OF THE LUNGS

- The extent to which the lungs will expand *for each unit increase* in trans-pulmonary pressure
- Lung compliance = $\frac{\text{Change in lung volume}}{\text{Change in TP pressure}}$
- Determination of compliance is useful as it is the *measure of stiffness of lungs*.
- Compliance is determined by two main factors

i. **Elastic forces of the lung tissue**-*one third of the total lung elasticity*

- ✓ *Emphysema and aging- Increase compliance*
- ✓ *Fibrosis- decrease compliance*

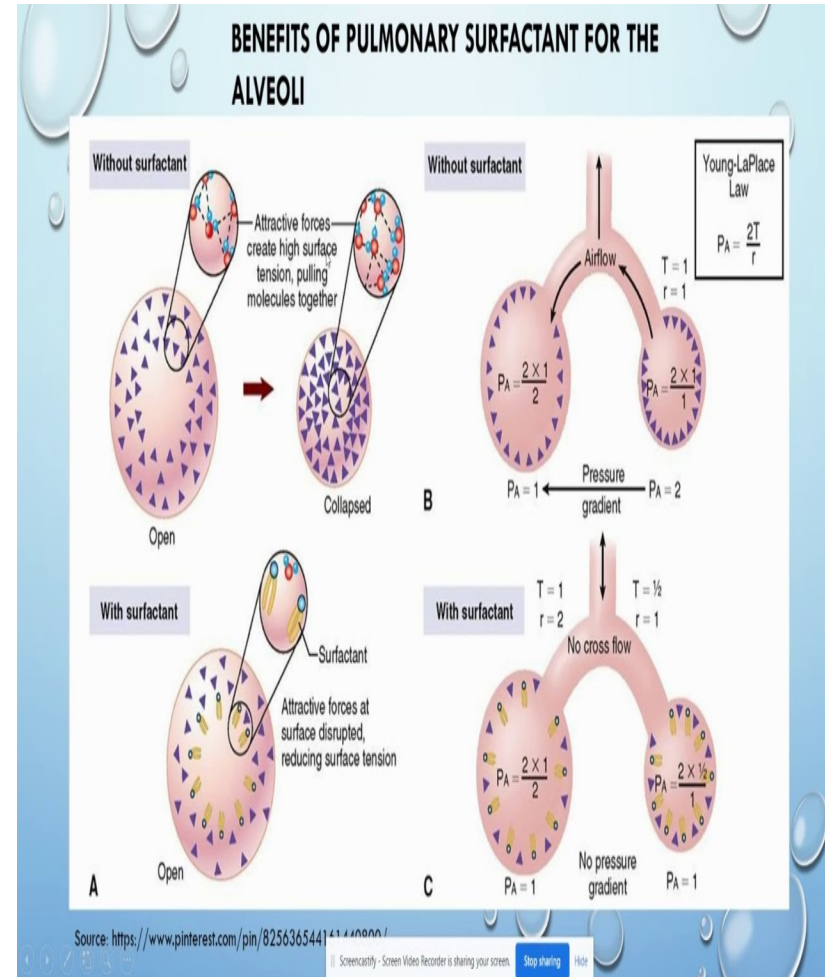
ii. **Elastic forces caused by surface tension**-*two third of the total lung elasticity*



Compliance of the lungs

ii. Surfactant and Its Effect on Surface Tension

- The most important components are the
 - ✓ Phospholipid-*dipalmitoyl phosphatidylcholine*
 - ✓ *surfactant apoproteins*, and
 - ✓ *calcium ions*
- Surface tension:
 - ✓ Pure water (**72 dynes/cm**)
 - ✓ Normal fluids lining the alveoli but without surfactant (**50 dynes/cm**)
 - ✓ Normal fluids lining the alveoli and *with normal amounts of surfactant* (**5 -30 dynes/cm**)



Q5: How surfactant reduce surface tension?

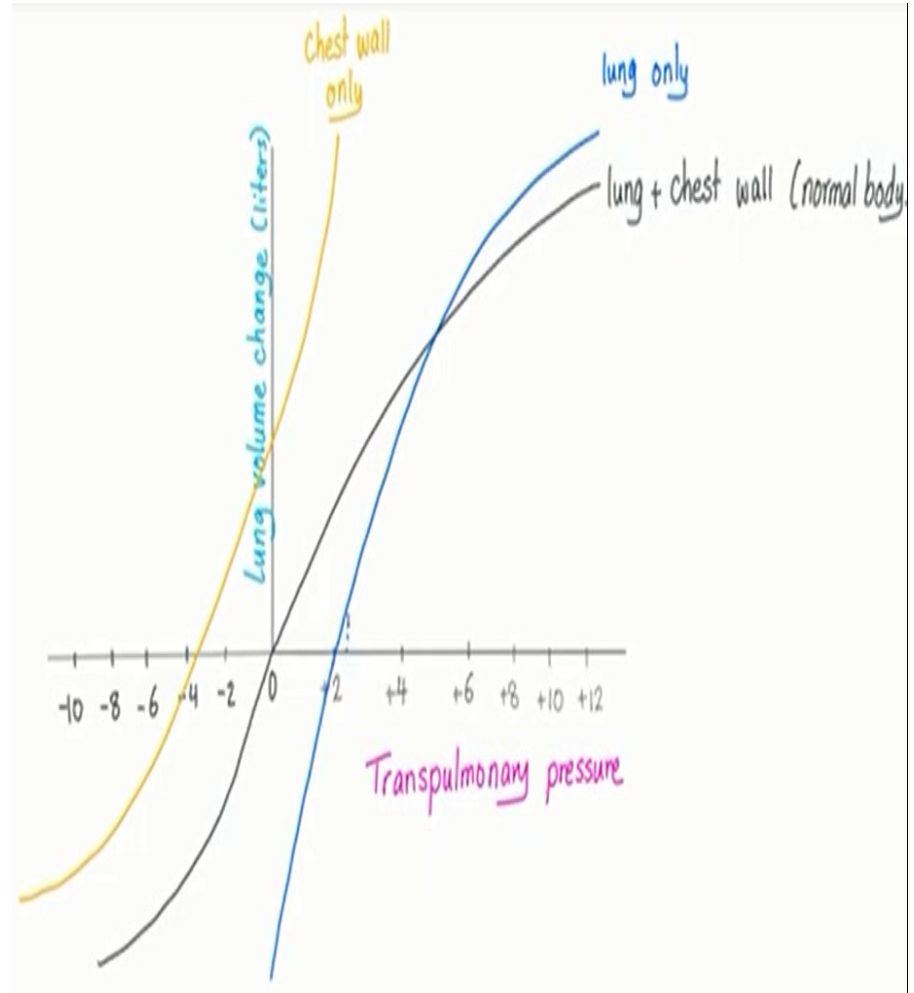
Compliance of the lungs

- Q6: Which of the following has highest compliance?

A. Lung alone

B. Chest wall alone

C. Lung + chest wall (Normal body)



Factors that diminished Lungs compliance

- Scar tissue or fibrosis that reduces the natural resilience of the lungs
- Blockage of the smaller respiratory passages with mucus or fluid
- Reduced production of surfactant (NRDS)
- Decreased flexibility of the thoracic cage or its decreased ability to expand

Pulmonary Function Tests

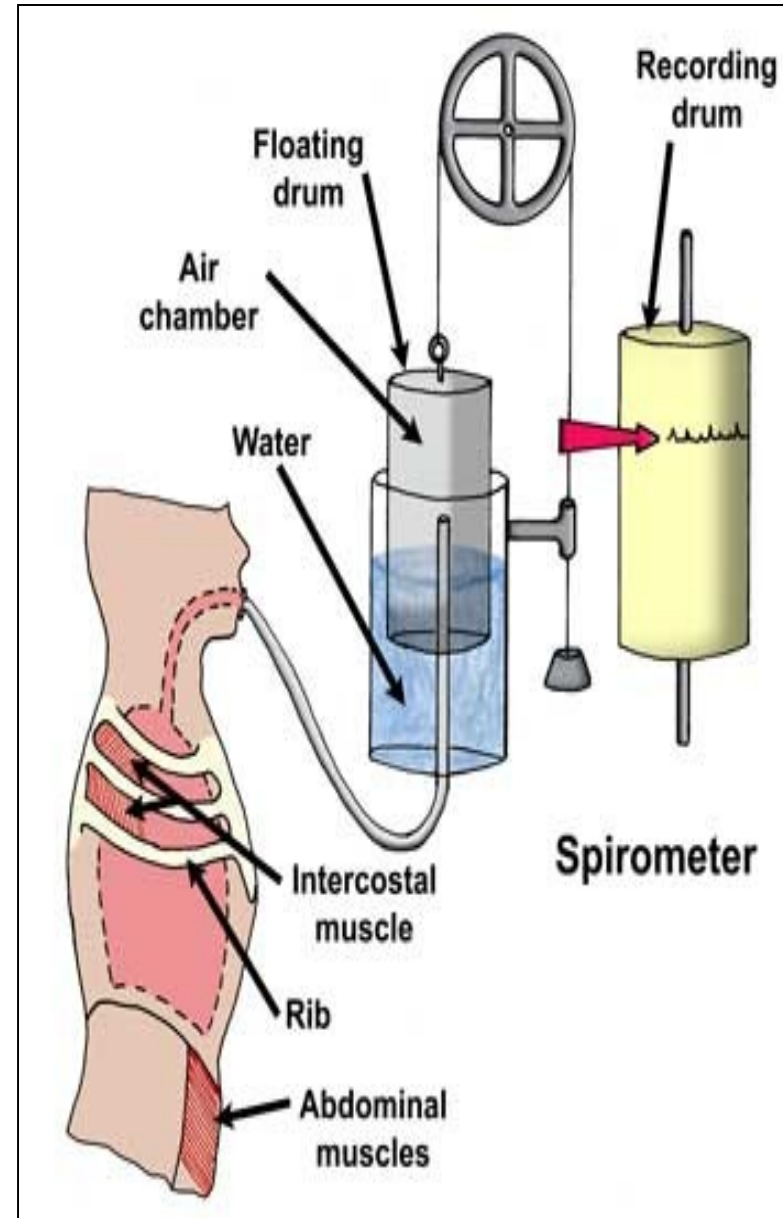
- ◆ The volume of air moved into and out of the lungs is measured using an apparatus spirometer
- ◆ An average individual has 14 breathing cycles in a minute and 6 -7 L of air circulates in the lung per minute

Spirometer

- ✓ an instrument consisting of a hollow bell inverted over water, used to evaluate respiratory function

Non-respiratory movements

- ✓ Mostly result from reflex action
- ✓ Examples include: coughing, sneezing, crying, laughing, hiccupping, and yawning



Pulmonary Function Tests...

There are 4 lung volumes and 4 lung capacities

1.Tidal volume (TV): volume of air that moves into and out of the lungs with each breath (approximately 500 ml)

2.Inspiratory reserve volume (IRV): volume of air that can be inspired forcefully beyond the tidal volume (3000 ml)

3.Expiratory reserve volume (ERV): volume of air that one can expire forcefully beyond the TV (1000 - 1200ml)

4.Residual volume (RV): volume of air left in the lungs after maximal expiration (1200 ml)

1.Inspiratory capacity (IC): total amount of air that a person can take in during maximal inspiration IC
$$= TV + IRV = 3300\text{ml} + 500\text{ l} = 3800\text{ ml}$$

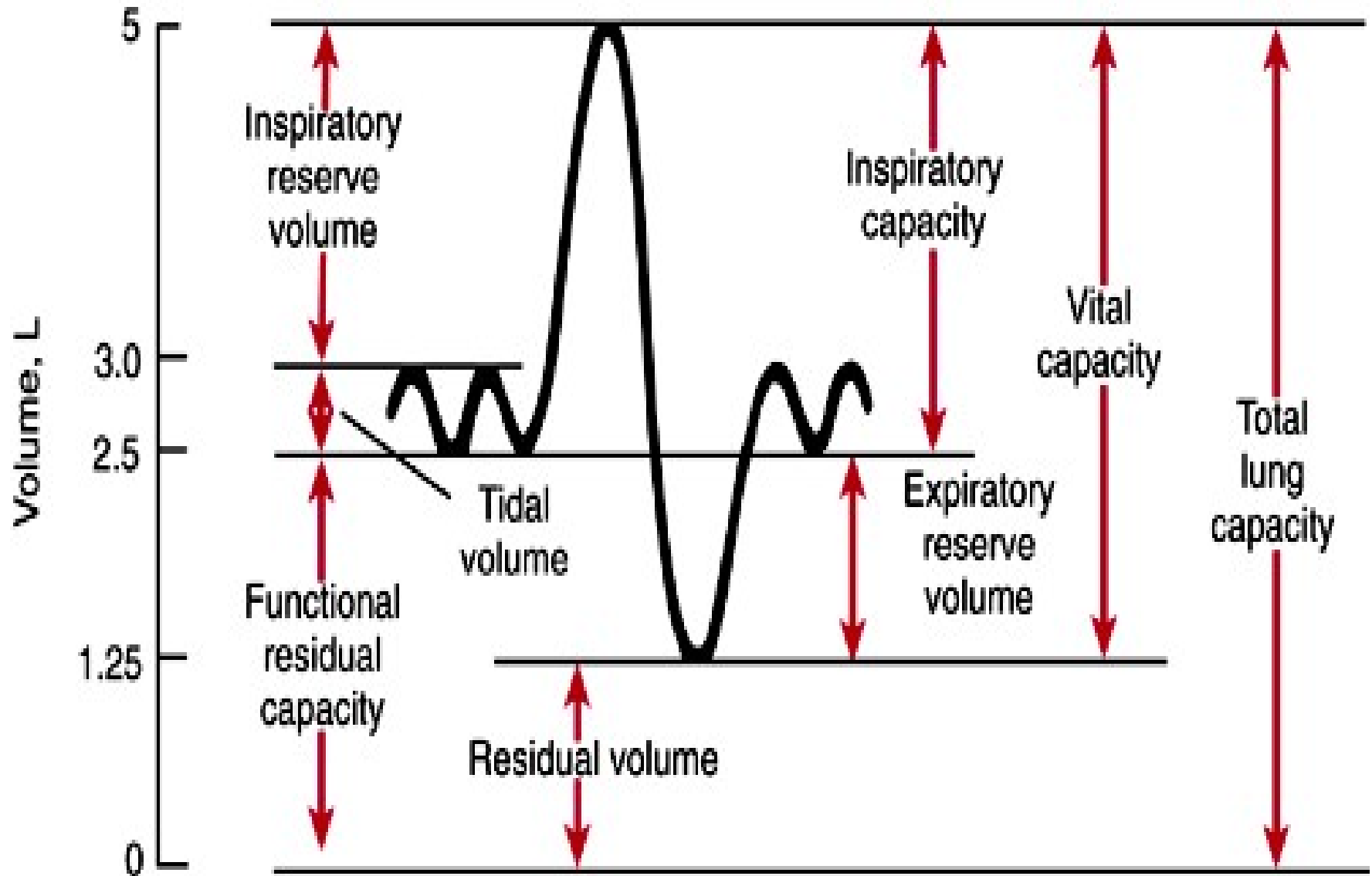
2.Functional residual capacity: amount of air remaining in the lungs at the end of normal expiration.
$$FRC = ERV + RV = 1000\text{ml} + 1200\text{ ml} = 2200\text{ml}$$

3.Vital capacity (VC): total amount of air that a person can expel from the lungs after maximal filling. VC
$$= TV + IRV + ERV = 500\text{ml} + 3300\text{ml} + 1000\text{ml} = 4800\text{ml}$$

4.Total lung capacity (TLC): maximum volume to w/c lungs can be expanded with maximal inspiratory effort.

Sum of all lung volumes (approximately in males is 6000 ml)

Pulmonary Function Tests...



Pulmonary ventilation rate /Minute Ventilation

- ◆ **Pulmonary Ventilation:** is the movement of air in and out of the lungs
- ◆ **pulmonary ventilation rate:** is defined as the tidal volume multiplied by the number of breaths taken per minute.
- ◆ Pulmonary ventilation rate depends upon 2 factors:
 - ✓ The size of each breath (tidal volume: TV)
 - ✓ The number of breaths/minute (respiratory Rate or frequency: RR)
$$PV = RR \times TV = 14 \times 500\text{ml} = 7000\text{ml/min}$$
 - ✓ Example: suppose your tidal volume is 500 mL (0.5 liters) and you breathe 15 times/minute:
 - ✓ Pulmonary ventilation rate = 15 breaths/min $\times$ 0.5 L/breath = 7.5 L/min

Pulmonary ventilation rate ...

- ◆ Breathing frequency at different age:
 - ✓ Adults = 10 – 20 breaths/min)
 - ✓ Children = 20 – 30 breaths/min
 - ✓ Infants = 30 – 40 breaths/min
 - ✓ New born = 40 – 50 breaths/min

Alveolar Ventilation Rate

- ◆ **Alveolar Ventilation:** how much air per minute enters the parts of the respiratory system in w/c gas exchange takes place
- ◆ **Dead Space:** volume of air filling the conducting zone and never contributes to gas exchange
- ◆ The AVR: Out of each breath approximately 150 cc is in the anatomical dead space (the conducting zone) where gas transport does not occur.
- ◆ In order to calculate the volume of air actually entering the alveoli the dead space volume must be subtracted.

$$✓ AVR = \text{Rate} \times [\text{Tidal volume} - 150 \text{ cc}]$$

Example:

$$AVR = 12 \times [500 - 150 \text{ cc}] =$$

$$AVR = 12 \times 350 \text{ cc} = 4200 \text{ cc}$$

Reading Assignment

Work of breathing

(Hint: Both normal person and in obstructive lung disease individuals)

Thank you